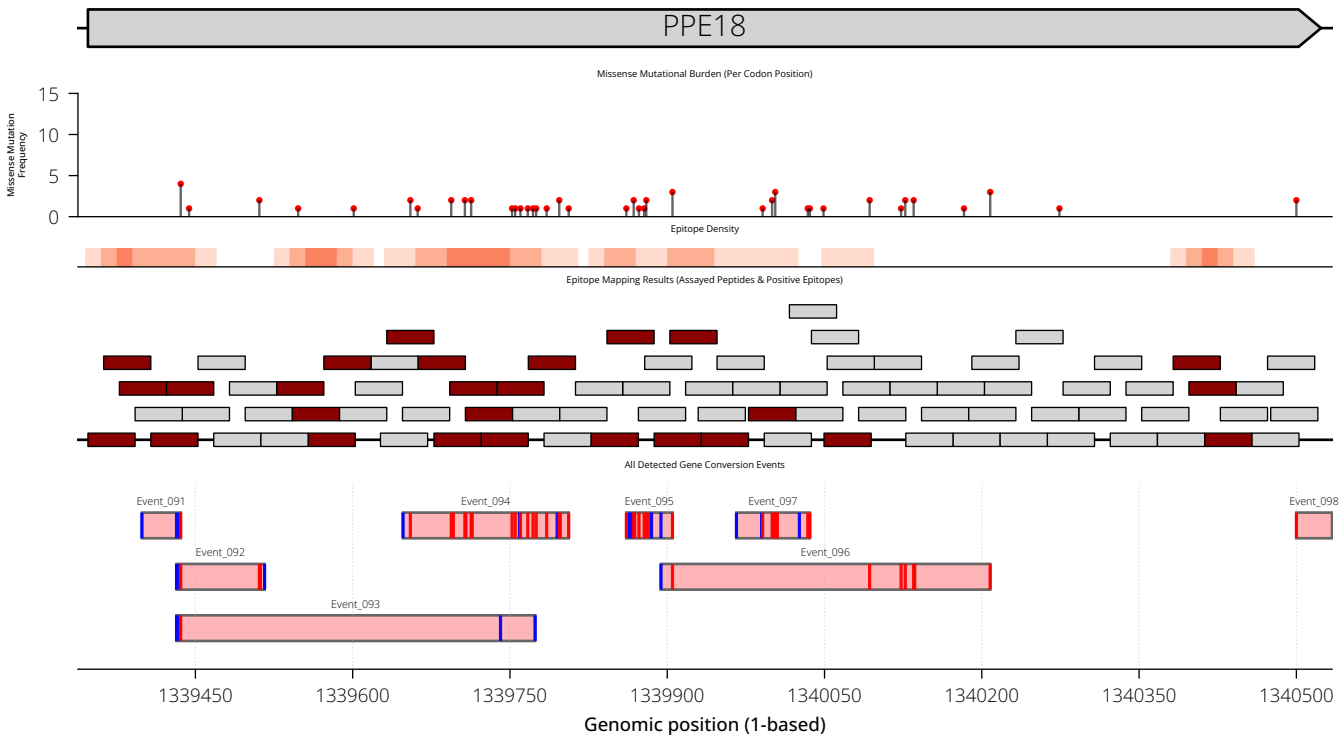
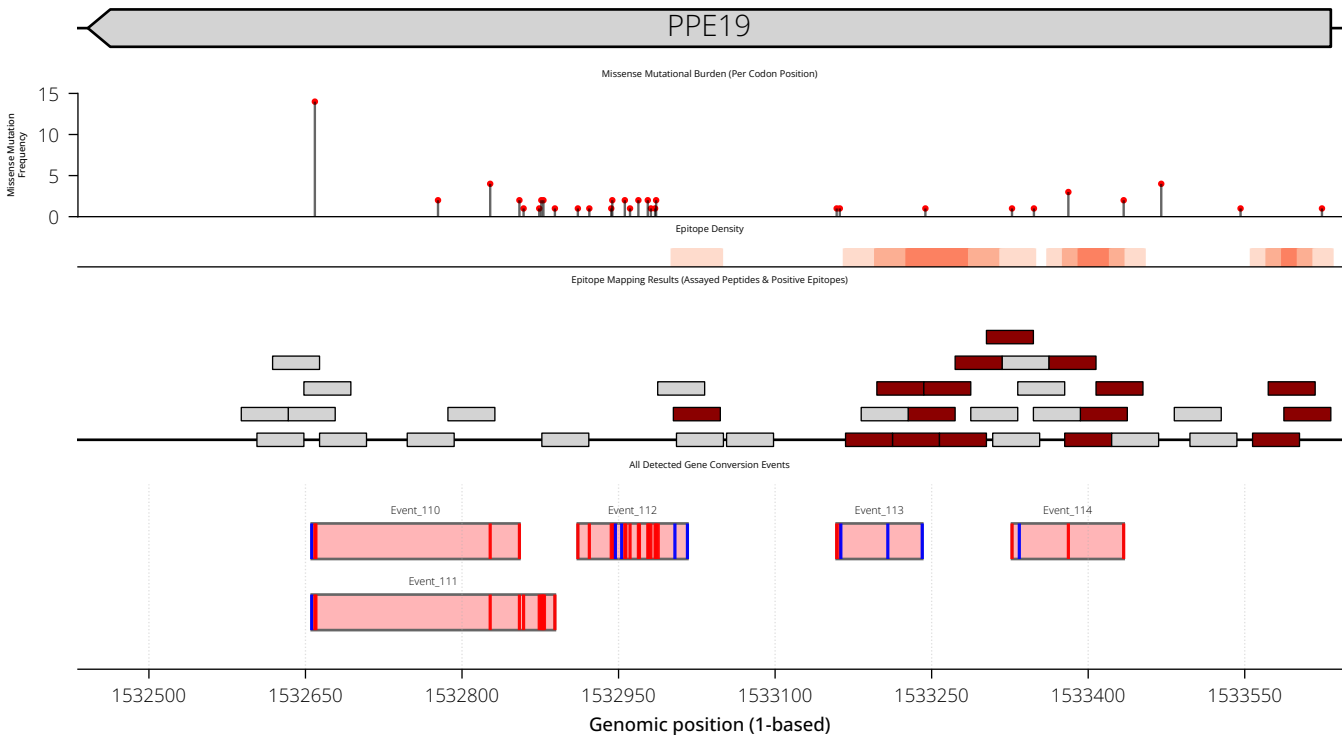


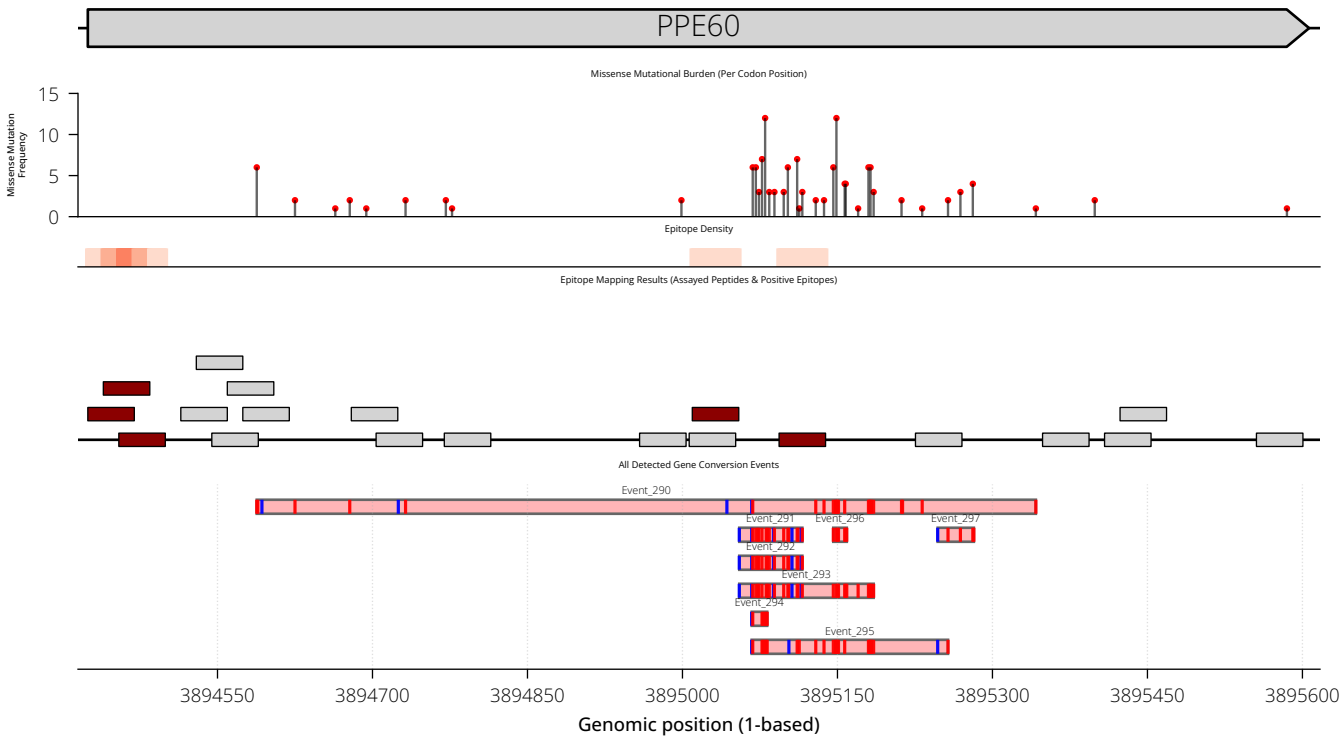
Gene(s): PPE18
 In Paralogous Region (PR)? : [True]
 # of Gene Conversion Events (GCE) detected: 8
 Missense Mutation Events Detected (Total): 61
 Missense Mutation Events Detected (In Epitopes): 44
 Total # of assayed peptides: 83
 # Positive T-cell epitopes: 27



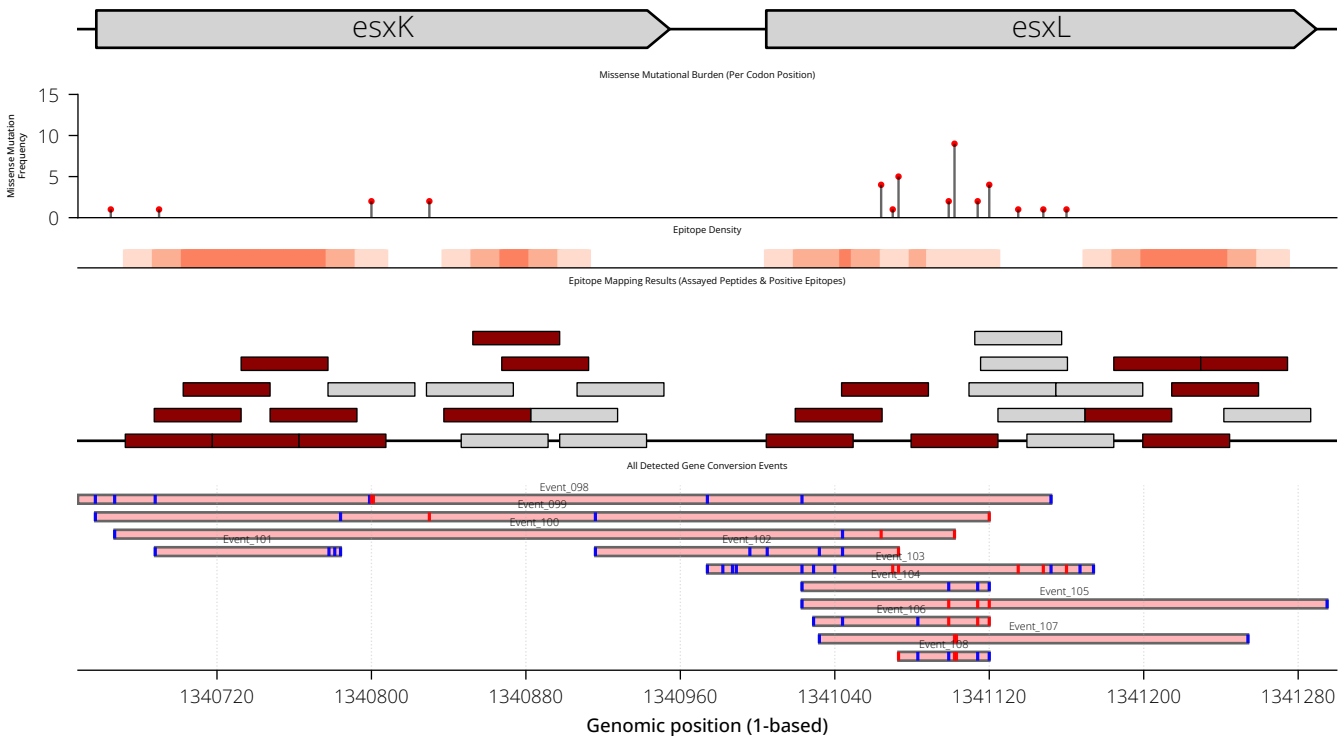
Gene(s): PPE19
In Paralogous Region (PR)? : [True]
of Gene Conversion Events (GCE) detected: 5
Missense Mutation Events Detected (Total): 61
Missense Mutation Events Detected (In Epitopes): 9
Total # of assayed peptides: 37
Positive T-cell epitopes: 16



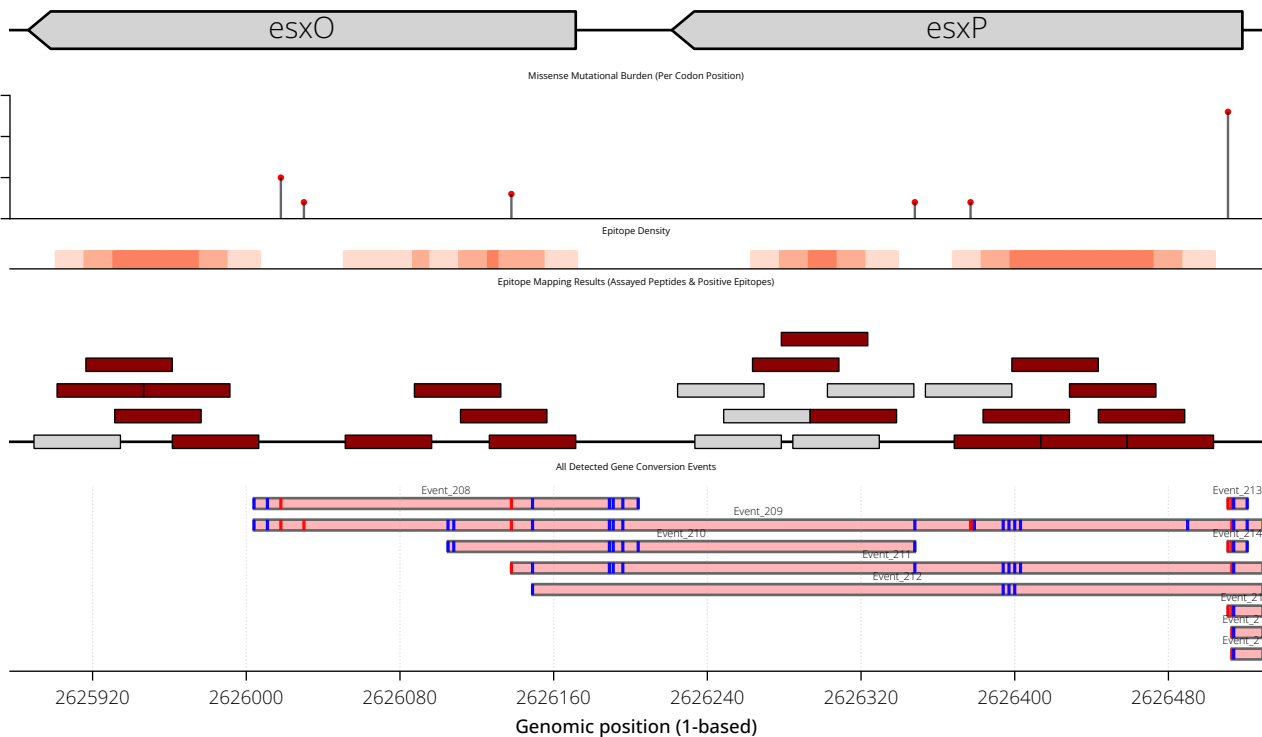
Gene(s): PPE60
In Paralogous Region (PR)? : [True]
of Gene Conversion Events (GCE) detected: 8
Missense Mutation Events Detected (Total): 141
Missense Mutation Events Detected (In Epitopes): 24
Total # of assayed peptides: 20
Positive T-cell epitopes: 5



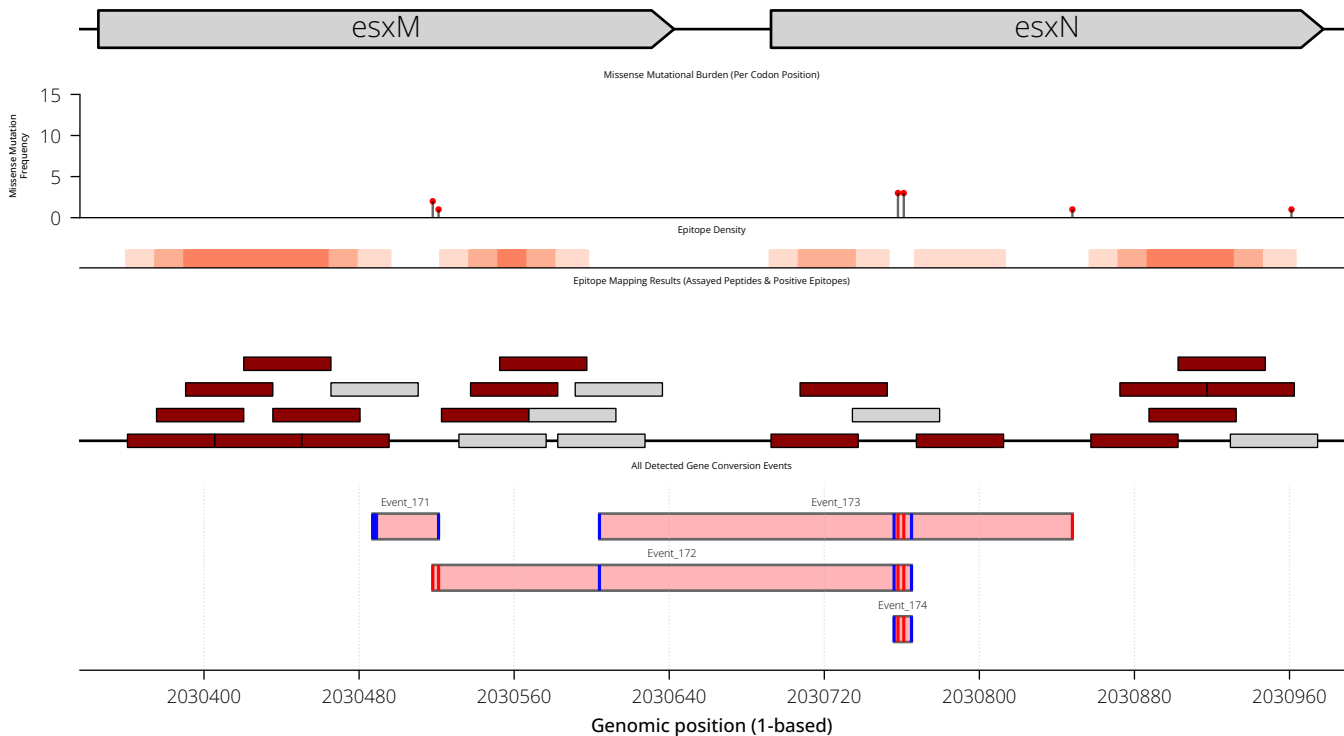
Gene(s): esxL,esxK
 In Paralogous Region (PR)? : [True True]
 # of Gene Conversion Events (GCE) detected: 15
 Missense Mutation Events Detected (Total): 36
 Missense Mutation Events Detected (In Epitopes): 30
 Total # of assayed peptides: 32
 # Positive T-cell epitopes: 19



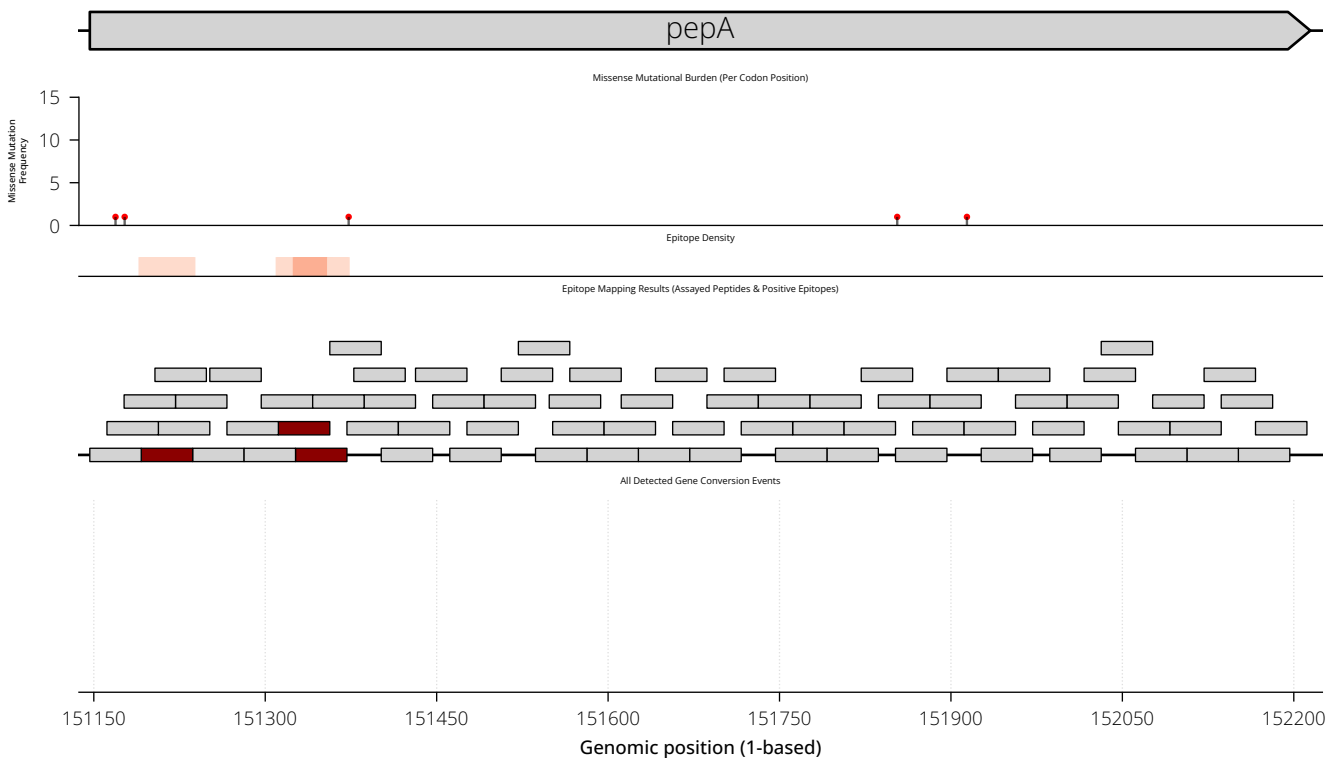
Gene(s): esxO,esxP
 In Paralogous Region (PR)? : [True True]
 # of Gene Conversion Events (GCE) detected: 14
 Missense Mutation Events Detected (Total): 27
 Missense Mutation Events Detected (In Epitopes): 5
 Total # of assayed peptides: 26
 # Positive T-cell epitopes: 19



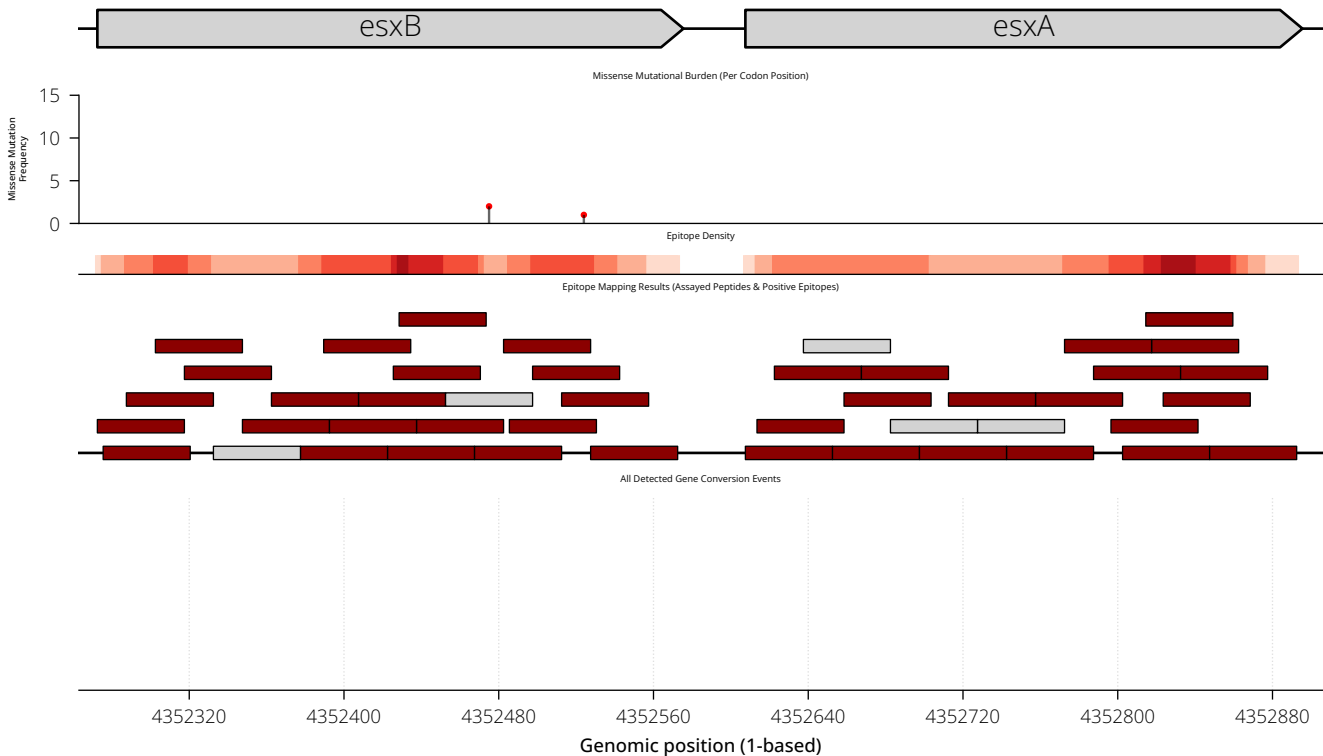
Gene(s): esxM,esxN
In Paralogous Region (PR)? : [True True]
of Gene Conversion Events (GCE) detected: 6
Missense Mutation Events Detected (Total): 11
Missense Mutation Events Detected (In Epitopes): 1
Total # of assayed peptides: 25
Positive T-cell epitopes: 18



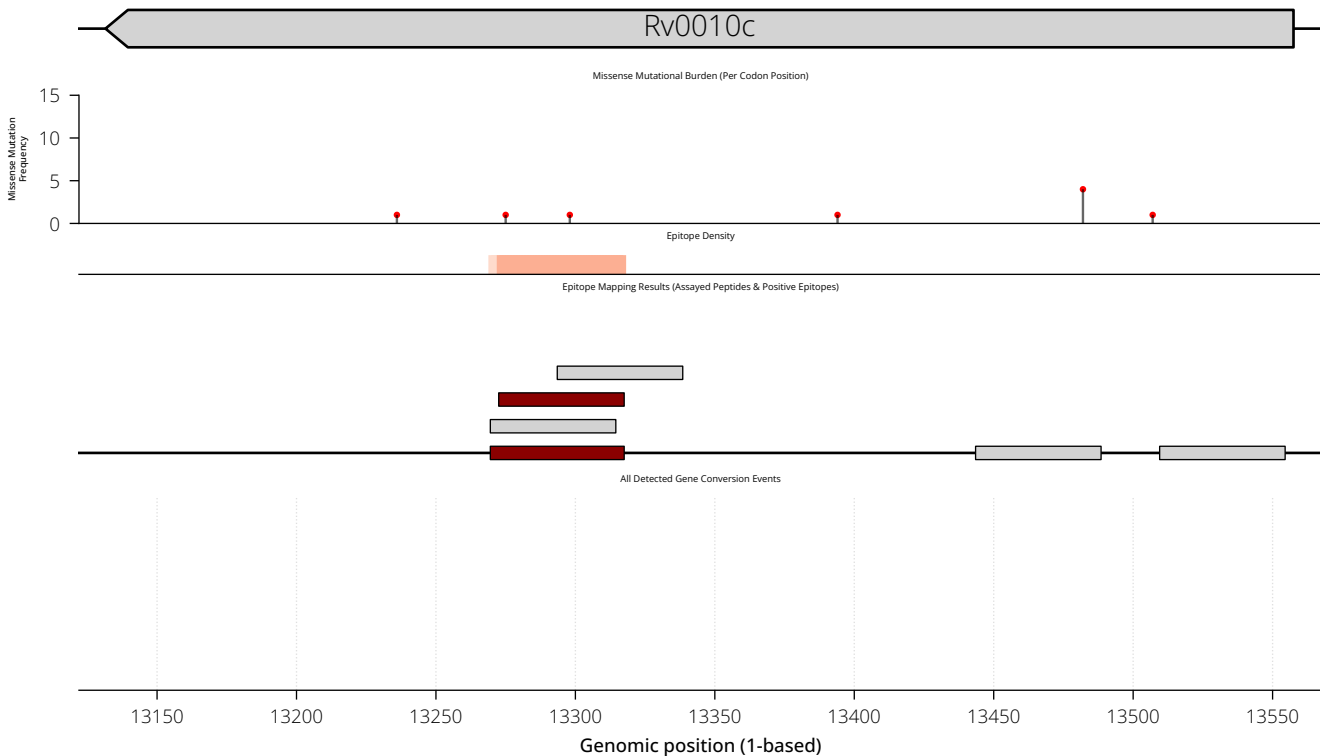
Gene(s): pepA
In Paralogous Region (PR)? : [False]
of Gene Conversion Events (GCE) detected: 0
Missense Mutation Events Detected (Total): 5
Missense Mutation Events Detected (In Epitopes): 0
Total # of assayed peptides: 72
Positive T-cell epitopes: 3



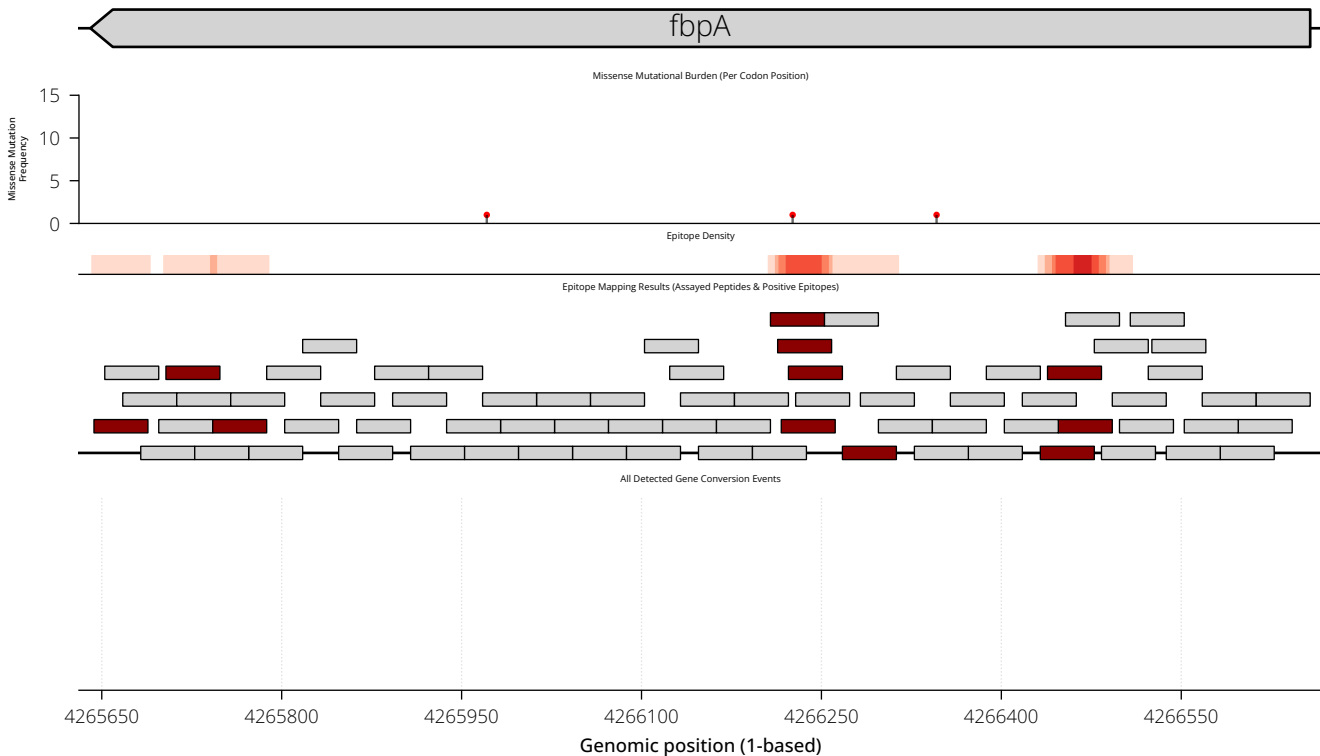
Gene(s): esxAesxB
 In Paralogous Region (PR)? : [False False]
 # of Gene Conversion Events (GCE) detected: 0
 Missense Mutation Events Detected (Total): 3
 Missense Mutation Events Detected (In Epitopes): 3
 Total # of assayed peptides: 45
 # Positive T-cell epitopes: 40



Gene(s): Rv0010c
In Paralogous Region (PR)? : [False]
of Gene Conversion Events (GCE) detected: 0
Missense Mutation Events Detected (Total): 9
Missense Mutation Events Detected (In Epitopes): 2
Total # of assayed peptides: 6
Positive T-cell epitopes: 2



Gene(s): fbpA
In Paralogous Region (PR)? : [False]
of Gene Conversion Events (GCE) detected: 0
Missense Mutation Events Detected (Total): 3
Missense Mutation Events Detected (In Epitopes): 1
Total # of assayed peptides: 78
Positive T-cell epitopes: 13



Gene(s): fbpB
In Paralogous Region (PR)? : [False]
of Gene Conversion Events (GCE) detected: 0
Missense Mutation Events Detected (Total): 3
Missense Mutation Events Detected (In Epitopes): 2
Total # of assayed peptides: 72
Positive T-cell epitopes: 16

